Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Next-Generation Targeted Protein Degradation Platform for the Treatment of Metabolic Syndrome**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio・Medical		Bio・Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Small-Molecule Pharmaceuticals	Medical Bio	Pharmaceutical Bio-based Technologies and Systems
Concept of Proposed Technology	Metabolic syndrome is a systemic disorder driven by impaired energy homeostasis across skeletal muscle, liver, adipose tissue, and bone, leading to insulin resistance, ectopic lipid accumulation, chronic inflammation, and increased cardiometabolic risk. Despite its rapidly increasing global prevalence and substantial clinical and socioeconomic burden, there are currently no disease-			

	modifying pharmacological therapies that directly address the underlying transcriptional and metabolic dysregulation of metabolic syndrome. Most existing therapeutic approaches focus on symptomatic control of individual components, resulting in limited durability and incomplete disease modification. The proposed technology introduces a mechanism-based therapeutic strategy targeting a central transcriptional regulator of systemic metabolic control: the PHF20–YY1 regulatory axis. PHF20 (PHD Finger Protein 20) functions as an epigenetic modulator influencing YY1-mediated repression of gene programs involved in mitochondrial function, energy utilization, and cellular metabolism, positioning it as a convergent regulatory node linking muscle metabolism, hepatic lipid handling, and bone homeostasis. Preclinical evidence supports the role of PHF20 as a negative regulator of metabolic efficiency in muscle-centric systems. While its therapeutic potential in broader metabolic disorders such as fatty liver disease and osteoporosis has not yet been fully explored, this biology highlights a clear opportunity for translational expansion through targeted intervention. Importantly, the proposed technology is built upon pre-existing proprietary assets, including (i) a validated PHF20 degradation–based biological mechanism established in muscle-centric systems, (ii) an in-house PHF20–YY1 reporter assay platform optimized for degrader discovery and optimization, and (iii) translational experimental systems capable of assessing metabolic and functional outcomes. While molecular glue–type PHF20 degraders optimized for metabolic syndrome indications have not yet been established, the foundational biology and screening infrastructure required for their development are already in place. This project focuses on the **biological and translational evaluation of PHF20 modulation** using existing chemical matter and validated mechanistic insights, rather than broad de novo medicinal chemistry expansion. A central objective of this program is to determine **how PHF20-centered mechanisms operate in metabolic syndrome–relevant disease contexts**, and to assess their functional impact on systemic energy metabolism. By interrogating a pathological transcriptional regulator in disease-relevant models, this approach aims to clarify the translational potential of PHF20 modulation for improving insulin sensitivity and muscle-driven metabolic homeostasis.
Rationale for Joint R&D Project	This joint R&D project is not intended for de novo target discovery or large-scale medicinal chemistry development, but rather to **evaluate and translate an already validated PHF20-centered biological mechanism into metabolic syndrome–relevant disease contexts** through focused global collaboration. While the role of PHF20 in muscle-centric metabolic regulation has been established, **its functional impact in metabolic syndrome remains to be clarified**, representing the primary objective of this program. The successful execution of this project requires the integration of complementary capabilities focused on **mechanistic biology and translational evaluation**, including: • Mechanistic analysis of PHF20–YY1–mediated metabolic regulation • Functional assessment of PHF20 modulation in metabolic disease–relevant systems • Translational in vivo and ex vivo models incorporating muscle-driven metabolic endpoints • Focused characterization of existing chemical matter to support biological interpretation These requirements cannot be efficiently addressed by a single organization alone. An international joint R&D framework enables the effective integration of mechanistic biology, translational disease modeling, and targeted chemical evaluation, aligning closely with the objectives of the TIPA International Joint R&D Program.
Key Words	1. Metabolic syndrome 2. Insulin resistance 3. Targeted Protein Degradation (TPD) 4. PHF20–YY1 axis 5. Muscle–metabolism crosstalk

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	**Overall Objective** To establish a first-in-class **therapeutic evaluation framework** for metabolic syndrome by validating how PHF20 modulation influences muscle-driven systemic metabolic homeostasis and by assessing its translational relevance in disease-relevant experimental models. **Specific Objectives** • **Characterize and evaluate existing PHF20-binding chemical matter** using validated biological and reporter-based assays to support targeted protein degradation–based mechanistic studies • **Assess the potential of PHF20 modulation**, including degradation-based approaches, to induce selective PHF20 downregulation in cellular systems • Validate the mechanism of action of PHF20 modulation through analysis of the PHF20–YY1 regulatory network and its impact on metabolic and mitochondrial gene programs • Evaluate functional and metabolic outcomes associated with PHF20 modulation using disease-relevant experimental models of metabolic syndrome, with a focus on muscle-driven metabolic parameters • Identify and prioritize **translationally relevant candidate molecules or approaches** suitable for subsequent development or co-development consideration			
R&D Target Specifications	The project targets the following **mechanistic and translational evaluation benchmarks**: • Demonstration of selective **PHF20 modulation**, including degradation where applicable, in cellular systems using reporter- and assay-based platforms • Confirmation of downstream YY1 repression and associated changes in metabolic and mitochondrial gene expression programs relevant to metabolic syndrome • Quantifiable changes in functional and metabolic endpoints, including muscle-associated metabolic parameters, in disease-relevant experimental models of metabolic syndrome • **Focused characterization** of potency, selectivity, and PHF20-modulating activity to support biological interpretation and translational assessment • Preliminary characterization of pharmacokinetic and safety-related properties of selected candidate molecules to inform translational relevance and future development decisions			
R&D Plan	The R&D program is structured into two integrated phases, progressing from mechanistic and functional evaluation of PHF20 modulation to translational validation in metabolic syndrome–relevant experimental models. **Phase 1: Hit Expansion and Advanced Mechanism Validation** • Application and refinement of an existing PHF20–YY1 reporter-based assay platform to evaluate PHF20 modulation and downstream pathway responses • Evaluation of focused sets of existing chemical matter informed by prior screening data and mechanistic hypotheses • Cellular assays to assess PHF20 modulation, YY1 suppression, and associated changes in metabolic and mitochondrial functional markers • Exploratory structure–activity relationship (SAR) analysis to support biological interpretation and prioritization of candidate chemical matter **Phase 2: Lead Optimization and In Vivo Validation** • Focused characterization of potency, selectivity, and PHF20-modulating activity of prioritized candidate molecules • Translational evaluation of candidate molecules in disease-relevant experimental models incorporating muscle-driven metabolic and functional endpoints • Preliminary characterization of pharmacokinetic and safety-related properties to inform translational relevance • Prioritization of translationally relevant candidate molecules or approaches to inform subsequent development or co-development decisions Milestones, go/no-go criteria, and quantitative benchmarks will be defined to support **mechanism-**			

	driven, development-informed, and risk-mitigated decision-making.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	These capabilities constitute essential prerequisites for effective execution of the proposed R&D scope, particularly for mechanistic and translational evaluation of PHF20 modulation. The research institute will focus on **mechanistic investigation and translational evaluation** required to assess the biological and functional relevance of PHF20 modulation in metabolic syndrome–relevant contexts, including: • Focused chemical characterization and structure-informed analysis, as needed, to support biological interpretation • Supportive evaluation of existing ligand information and chemical matter to inform mechanistic and translational studies • Mechanistic validation of PHF20–YY1–mediated regulatory pathways and associated protein–protein interaction dynamics • Support for chemical handling, formulation considerations, and focused characterization required for translational evaluation
Expected R&R of Korean SME Partner	The Korean SME partner is expected to possess proprietary, internally developed **biological and translational evaluation platforms** relevant to PHF20 modulation and metabolic disease contexts, including: • PHF20-specific modulation and degradation assays, together with PHF20–YY1 reporter systems, enabling mechanism-based biological evaluation • Translational experimental platforms capable of assessing **muscle-driven metabolic regulation and functional outcomes** in disease-relevant contexts • Prior experience advancing muscle- and metabolism-related therapeutic programs through **late preclinical and translational development stages** These capabilities constitute essential prerequisites for effective execution of the proposed R&D scope, particularly for biological validation and translational assessment. The Korean SME partner will lead **biological validation and translational evaluation activities**, including: • Core biological validation of the PHF20–YY1 regulatory axis in **muscle-centric and metabolic disease–relevant contexts** • Operation of proprietary HTS and reporter assay platforms supporting **mechanism-based evaluation of PHF20 modulation** • Translational functional evaluation using disease-relevant experimental models incorporating **metabolic and muscle-associated endpoints relevant to metabolic syndrome** • Integration of pharmacology, DMPK, and translational datasets to support **mechanism-informed and development-oriented** decision-making

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Outcomes** • Establishment of PHF20 as a **therapeutically relevant transcriptional regulator** implicated in systemic metabolic dysregulation through mechanistic and translational validation • **Evaluation and validation of PHF20-targeted modulation strategies**, including degradation-based approaches where applicable, through global collaborative R&D • Creation of a **validated mechanistic and translational evaluation framework** applicable to metabolic syndrome and muscle-driven metabolic disorders **Commercial and Industrial Impact** • Strong licensing or co-development potential with global pharmaceutical partners seeking **mechanism-validated therapeutic strategies** for metabolic diseases • Strategic expansion opportunities into related metabolic indications, including fatty liver disease, insulin resistance–associated disorders, and sarcopenic obesity, **based on validated mechanistic insights** • Long-term pipeline value through **platform-based therapeutic evaluation approaches** applicable to metabolic and age-related diseases **Societal Impact** • Potential reduction in long-term healthcare costs associated with metabolic syndrome and its complications • Improvement in quality of life through enhanced metabolic health and preservation of muscle-driven functional capacity • Strengthening of international competitiveness of Korean biotechnology innovation through **focused global joint R&D and mechanism-based translational research**

6. References

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